

```
In [ ]: import pandas as pd
df=pd.read_csv("/content/heart.csv")
print(df.head())
print(df.tail())
print(df.info())
print(df.describe())
print(df.columns)
```

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS
RestingECG						
0	40	M	ATA	140	289	0
Normal						
1	49	F	NAP	160	180	0
Normal						
2	37	M	ATA	130	283	0
ST						
3	48	F	ASY	138	214	0
Normal						
4	54	M	NAP	150	195	0
Normal						

	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	N	0.0	Up	0
1	N	1.0	Flat	1
2	N	0.0	Up	0
3	Y	1.5	Flat	1
4	N	0.0	Up	0

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS
RestingECG						
913	45	M	TA	110	264	0
Normal						
914	68	M	ASY	144	193	1
Normal						
915	57	M	ASY	130	131	0
Normal						
916	57	F	ATA	130	236	0
LVH						
917	38	M	NAP	138	175	0
Normal						

	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
913	132	N	1.2	Flat	1
914	141	N	3.4	Flat	1
915	115	Y	1.2	Flat	1
916	174	N	0.0	Flat	1
917	173	N	0.0	Up	0

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 918 entries, 0 to 917

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	Age	918 non-null	int64
1	Sex	918 non-null	object

```

2  ChestPainType    918 non-null    object
3  RestingBP        918 non-null    int64
4  Cholesterol      918 non-null    int64
5  FastingBS        918 non-null    int64
6  RestingECG       918 non-null    object
7  MaxHR            918 non-null    int64
8  ExerciseAngina   918 non-null    object
9  Oldpeak          918 non-null    float64
10 ST_Slope         918 non-null    object
11 HeartDisease     918 non-null    int64

```

```
dtypes: float64(1), int64(6), object(5)
```

```
memory usage: 86.2+ KB
```

```
None
```

	Age	RestingBP	Cholesterol	FastingBS	MaxHR
\					
count	918.000000	918.000000	918.000000	918.000000	918.000000
mean	53.510893	132.396514	198.799564	0.233115	136.809368
std	9.432617	18.514154	109.384145	0.423046	25.460334
min	28.000000	0.000000	0.000000	0.000000	60.000000
25%	47.000000	120.000000	173.250000	0.000000	120.000000
50%	54.000000	130.000000	223.000000	0.000000	138.000000
75%	60.000000	140.000000	267.000000	0.000000	156.000000
max	77.000000	200.000000	603.000000	1.000000	202.000000

	Oldpeak	HeartDisease
count	918.000000	918.000000
mean	0.887364	0.553377
std	1.066570	0.497414
min	-2.600000	0.000000
25%	0.000000	0.000000
50%	0.600000	1.000000
75%	1.500000	1.000000
max	6.200000	1.000000

```

Index(['Age', 'Sex', 'ChestPainType', 'RestingBP', 'Cholesterol',
      'FastingBS',
      'RestingECG', 'MaxHR', 'ExerciseAngina', 'Oldpeak',
      'ST_Slope',
      'HeartDisease'],
      dtype='object')

```

```

In [ ]: categorical_cols = ['Sex', 'ChestPainType', 'RestingECG',
                          'ExerciseAngina', 'ST_Slope']
for col in categorical_cols:
    unique_vals = df[col].unique()
    print(f"\nUnique values in '{col}': {unique_vals}")

```

```
Unique values in 'Sex': ['M' 'F']
```

```
Unique values in 'ChestPainType': ['ATA' 'NAP' 'ASY' 'TA']
```

```
Unique values in 'RestingECG': ['Normal' 'ST' 'LVH']
```

```
Unique values in 'ExerciseAngina': ['N' 'Y']
```

```
Unique values in 'ST_Slope': ['Up' 'Flat' 'Down']
```

```
In [ ]: df['ExerciseAngina'] = df['ExerciseAngina'].replace('3+', 3)
df['ExerciseAngina'] =
pd.to_numeric(df['ExerciseAngina'],errors='coerce').astype('Int64')
print("\nValue counts in 'ExerciseAngina' after conversion:")
print(df['ExerciseAngina'].value_counts(dropna=False))
```

Value counts in 'ExerciseAngina' after conversion:

ExerciseAngina

<NA> 918

Name: count, dtype: Int64

```
In [ ]: cols = ['Age', 'RestingBP', 'Cholesterol', 'FastingBS', 'MaxHR',
               'Oldpeak']

for col in cols:
    print(f"\nStatistics for '{col}':")
    print(f"Mean: {df[col].mean():.2f}")
    print(f"Median: {df[col].median():.2f}")
    print(f"Mode: {df[col].mode().values}")
    print(f"Range: {df[col].max() - df[col].min():.2f}")
    print(f"Variance: {df[col].var():.2f}")
    print(f"Standard Deviation: {df[col].std():.2f}")
```

```
Statistics for 'Age':
Mean: 53.51
Median: 54.00
Mode: [54]
Range: 49.00
Variance: 88.97
Standard Deviation: 9.43
```

```
Statistics for 'RestingBP':
Mean: 132.40
Median: 130.00
Mode: [120]
Range: 200.00
Variance: 342.77
Standard Deviation: 18.51
```

```
Statistics for 'Cholesterol':
Mean: 198.80
Median: 223.00
Mode: [0]
Range: 603.00
Variance: 11964.89
Standard Deviation: 109.38
```

```
Statistics for 'FastingBS':
Mean: 0.23
Median: 0.00
Mode: [0]
Range: 1.00
Variance: 0.18
Standard Deviation: 0.42
```

```
Statistics for 'MaxHR':
Mean: 136.81
Median: 138.00
Mode: [150]
Range: 142.00
Variance: 648.23
Standard Deviation: 25.46
```

```
Statistics for 'Oldpeak':
Mean: 0.89
Median: 0.60
```

Mode: [0.]

Range: 8.80

Variance: 1.14

Standard Deviation: 1.07

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